Performance measurements of model

1) Classification →

2) Regression

$M() \to$ good ? → Acc ↑

x	y	predicted
x1	1	0
x2	1	0

train → $\boxed{f()}$ → judge → $\boxed{Acc}$

(x, y)

KNN

$\frac{0}{2}$ → Acc

1) Acc

↳ $\frac{\text{\# of correctly classified pts}}{\text{total \# pts in } D_{Test}}$

$0 \le acc \le 1$

(0 - 100)

BEST

worst

eg 100pts

→ 60 +ve → 53 +ve → 7 time → -ve

→ 40 -ve → 35 -ve → 5 time → +ve

error → 12%

correctly → 88%

acc = 88%

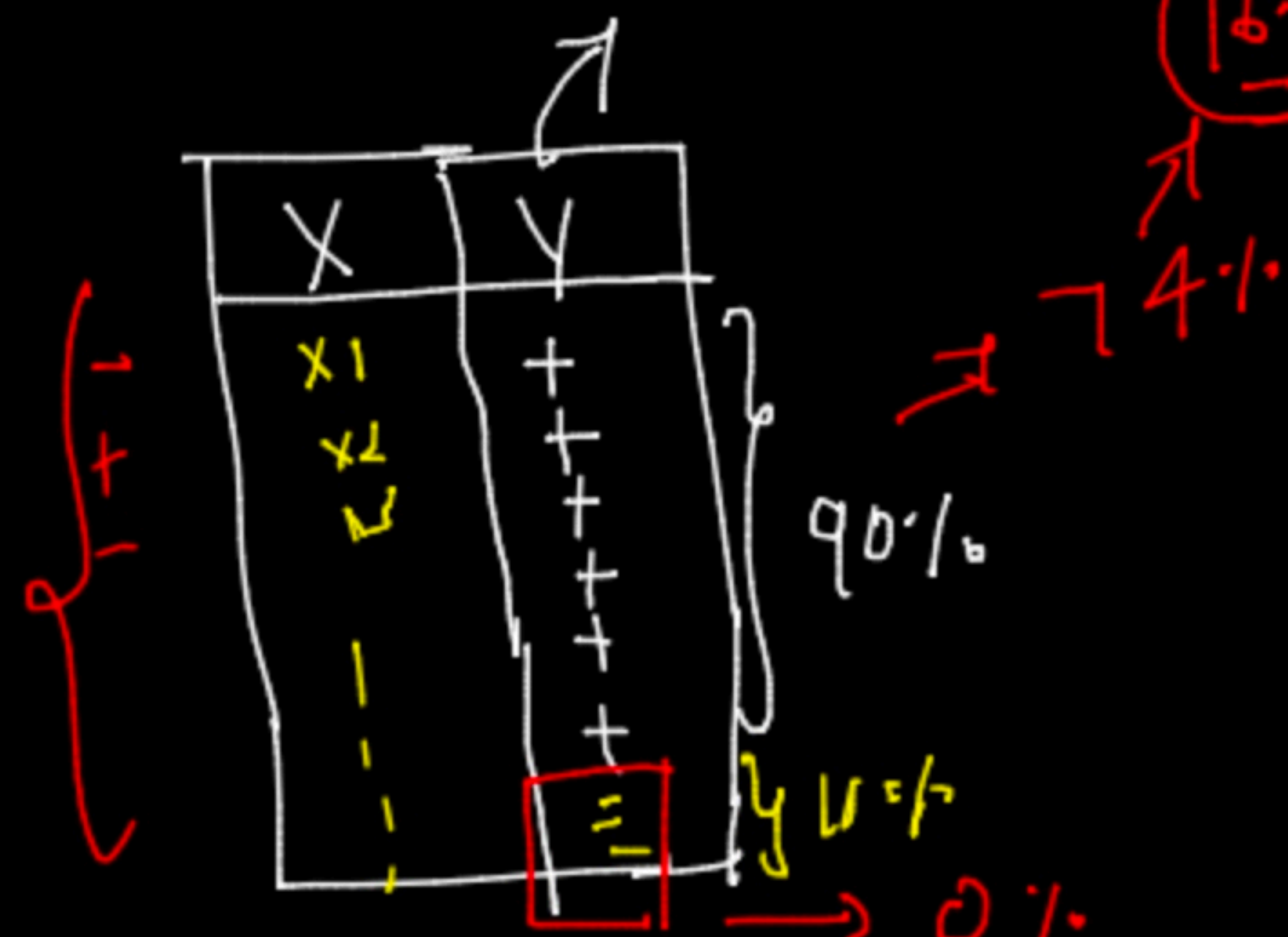
Problem with Accu

(i) imbalance data

interview

Orest

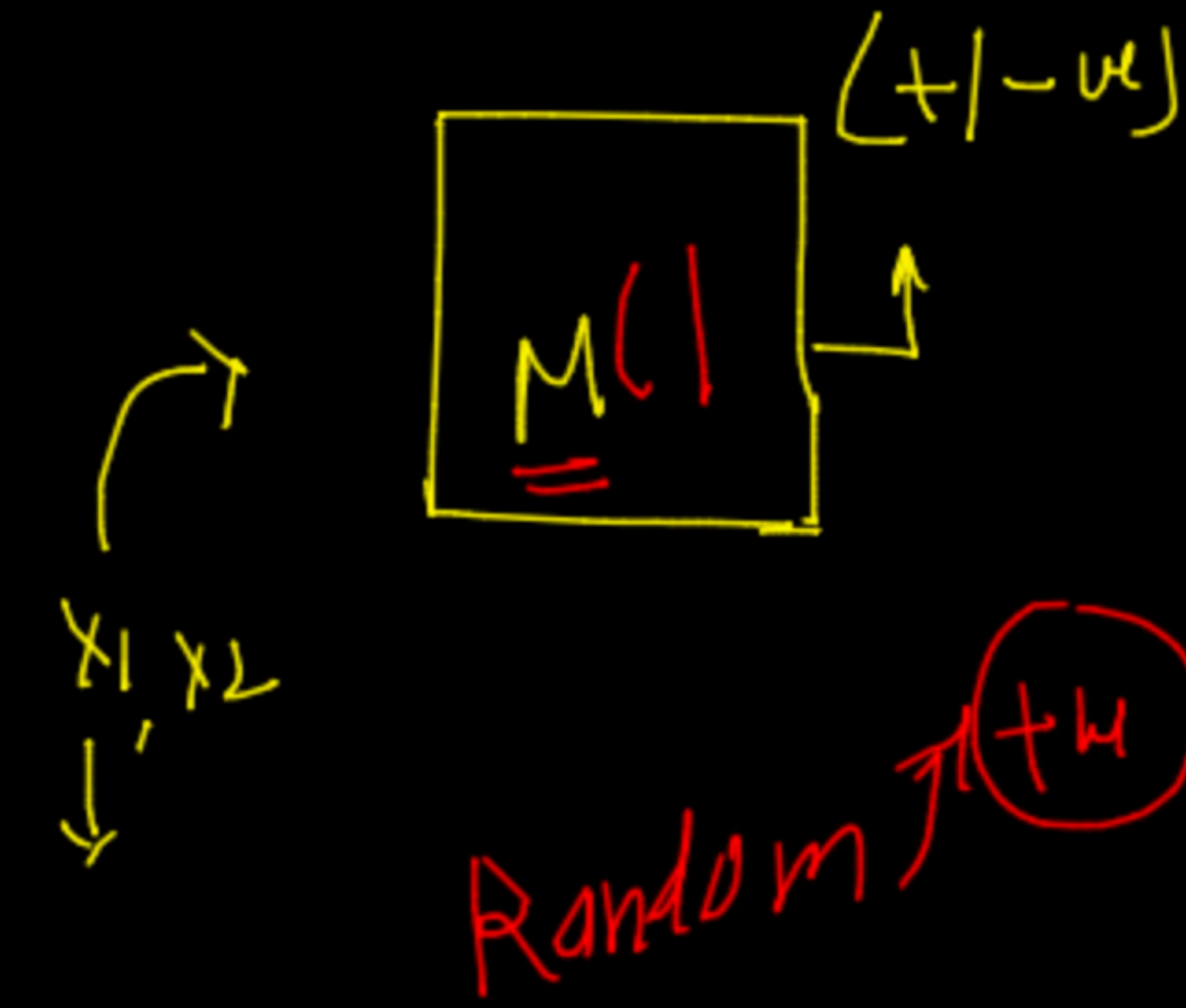
(+ve/-ve)



-10%

90% time

{ 74%



90% time

No

(ii)

X	Y	M1	M2	$\hat{Y}_1$	$\hat{Y}_2$
x1	1 ✓	0.9	0.6	1 ✓	1
x2	1 ✓	0.8	0.65	1 ✓	1
x3	0	0.1	0.15	0	0
x4	0	0.15	0.48	0	0

$\frac{4}{4} \rightarrow 100\%$

But?

same

conclusion

prob-score → Acc ↑

$\boxed{M1}$ → BEST

$\frac{4}{4} \rightarrow 100\%$

if prob-score > 50%

class → 1

Rule

x1

$\boxed{M_1(\ )}$

(KNN)

$\boxed{M_2(\ )}$

D_{Train}

1. Confusion-Matrix

↳ classification $Y \in \{+ve, -ve\}$

↑ ↑

Binary classification [Y] ✓

'Actual'

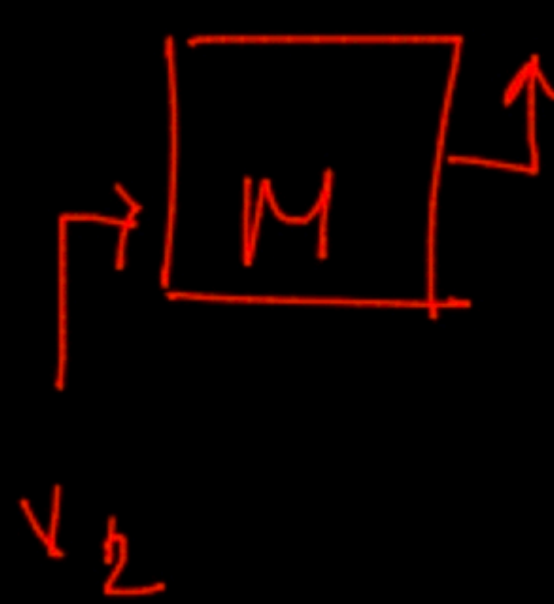
Predicted	0	1
0	2 $\cancel{a}$	$\cancel{b}$ 1
1	1 $\cancel{c}$	$\cancel{d}$ 1

Actual

Test

X	Y	$\hat{Y}$
X1	0	0
X2	0	0
X3	0	1
X4	1	1
X5	1	0

⇐ o/p



a ->

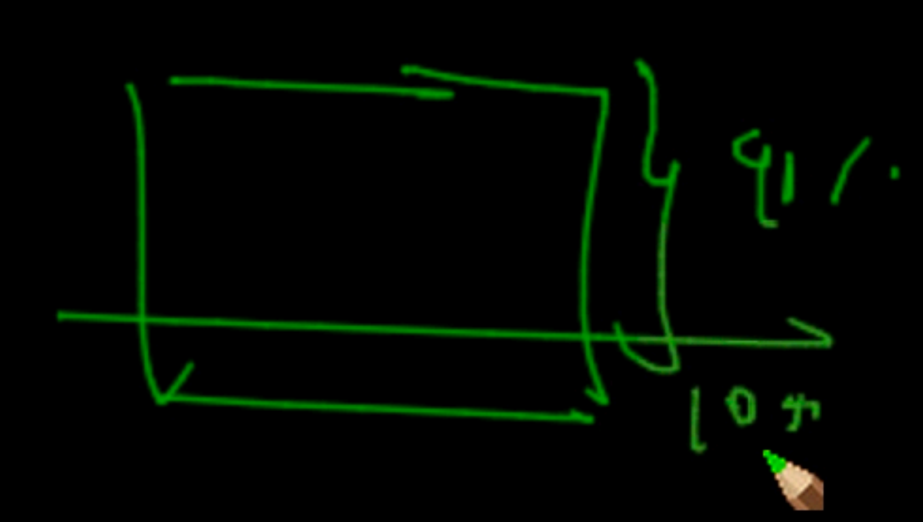
How many points which have predicted value as '0' and actual value is also '0' case 1

	0	1
0	2	0
1	0	2

↳ good model

	y	$\hat{y}$
X1	0	0
X2	0	0
X3	1	1
X4	1	1

imbalance data



	0	1
0	75 ↑	2 ↓
1	5 ↓	8 ↑

⇓ good

100 pts → 90 → +ve →

↳ 10 (-ve) < 8

Actual

predicted

	0	1
0	TN	FN
1	FP	TP

total pts in your data $= N + P$

good model

(i) $TPR = \frac{\#TP}{P} \rightarrow \frac{94}{100} \rightarrow 94\%$

(ii) $TNR = \frac{850}{900} \Rightarrow =$

e.g.

	0	1
0	850 TN	6 FN
1	50 FP	94 TP

$\underset{N}{900} + \underset{P}{100} \Rightarrow 1000$

Testing

T P

are you correct?

what is predicted label

(iii) $FPR = \frac{50}{900} \Rightarrow \downarrow$

(iv) $FNR = \frac{6}{100} \Rightarrow 6\% \downarrow$

→ Precision, Recall and F1-score

	0	1
0	TN	FN
1	FP	TP
	N	P

Recall

↳ $TPR = \frac{TP}{P}$

$$F1\text{-score} = 2 * \left(\frac{Pr * Re}{Pr + Recall} \right)$$

Pr →

$$\frac{TP}{TP + FP}$$

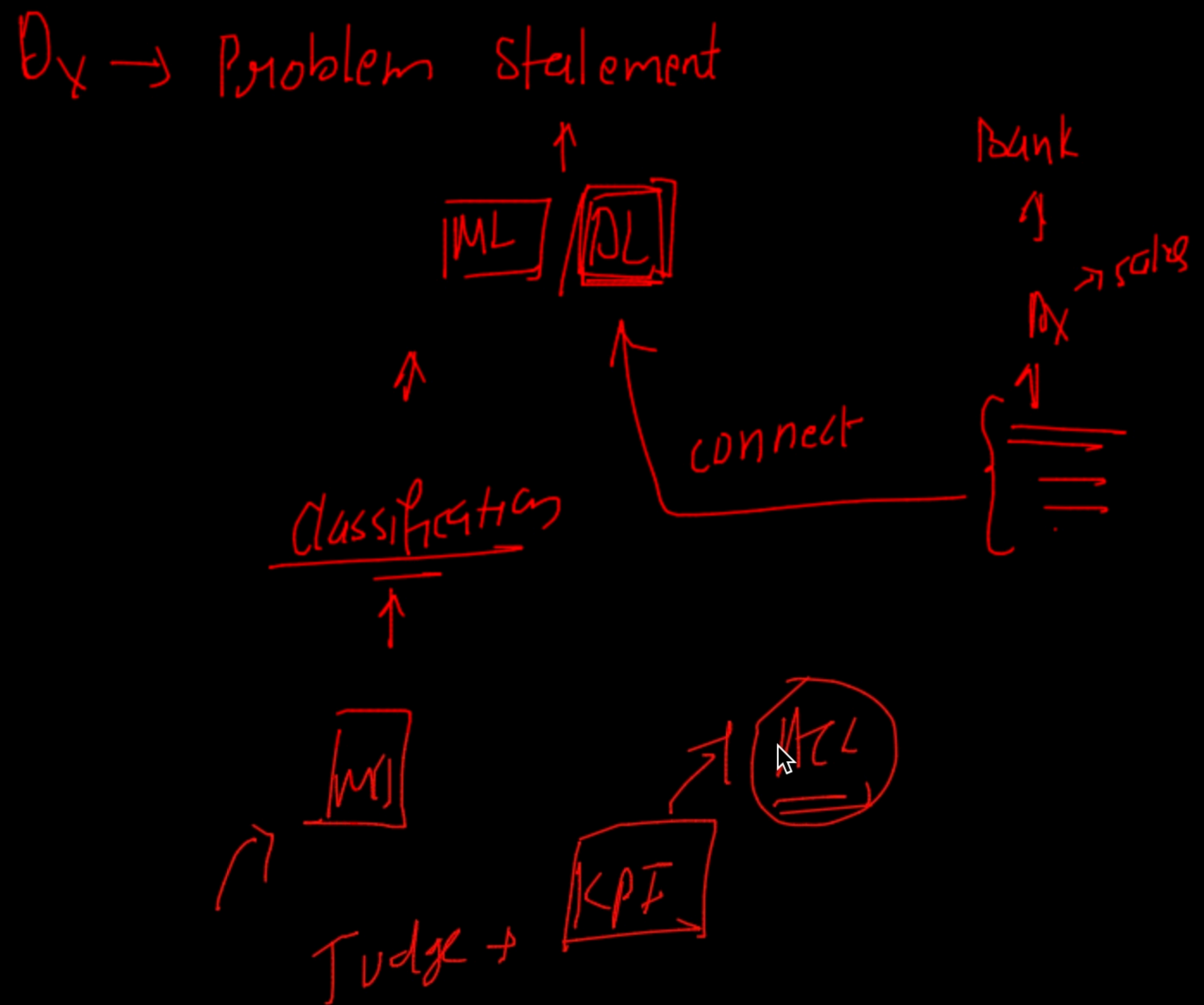
→ of all the pts the model declared/predicted to have +ve | what %age of them are actually +ve

imbalance ↓ accx →

Pre ↑ Recall ↑

F1-score (0-1)

confusion matrix


Dx → Problem Statement
ML
DL
Bank
sales
connect
classification
MRI
HCL
KPI
Judge →

ROC and AUC

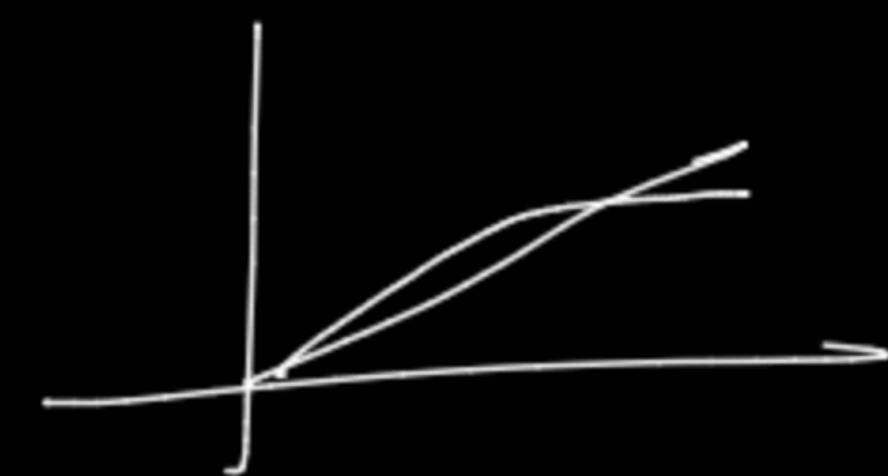
Area under curve

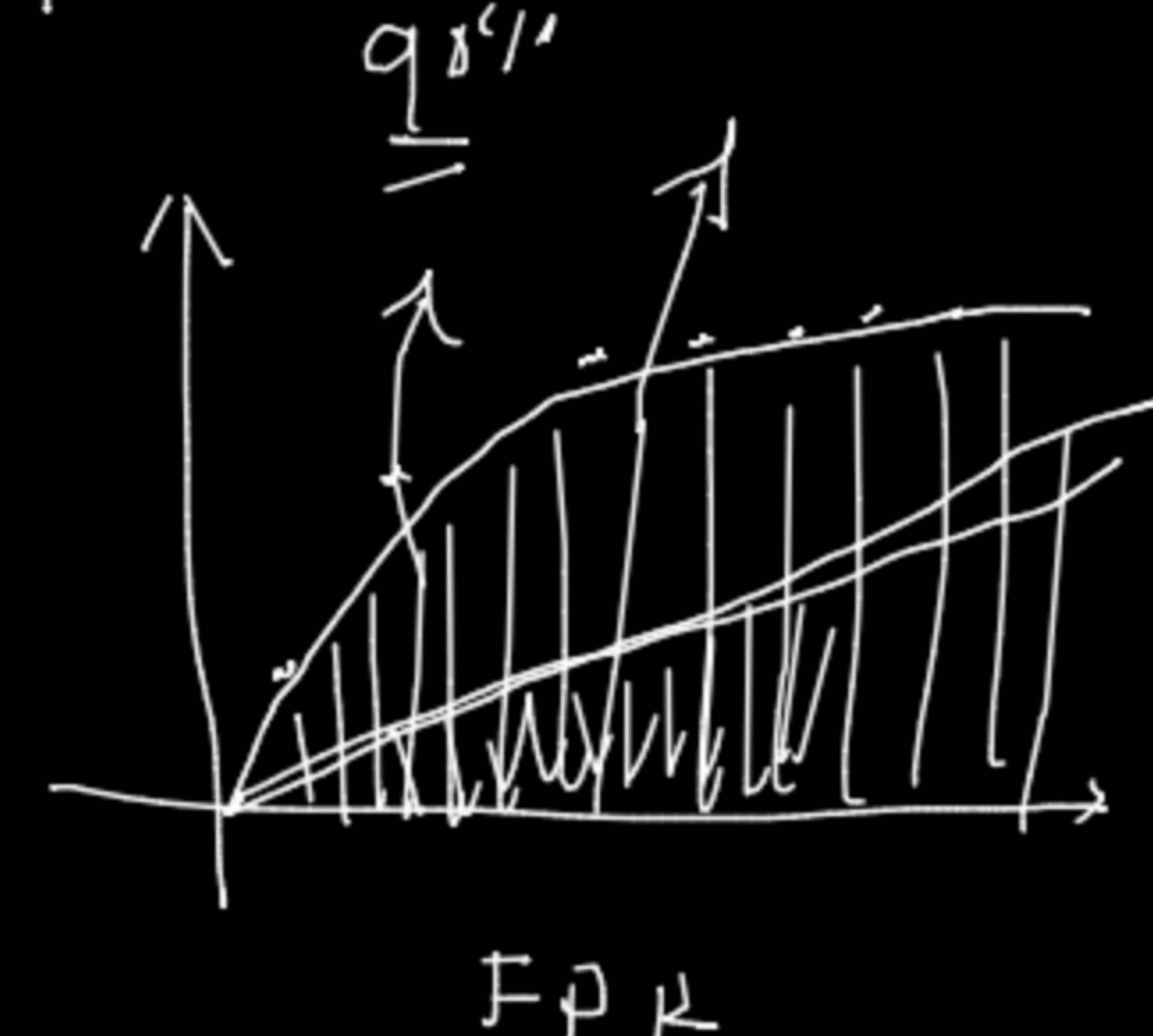
Receiver operating characteristics → 2^{n9}

↳ Binary classification

Probability $\geq 0.95 \rightarrow$ 1
0

$\geq 0.92 \rightarrow 1$





In code

Step 1

sort the $\hat{y}$ in decreasing order →

X	Y	$\hat{Y}$	$\hat{Y}_{\geq 0.95}$	$\hat{Y}_{0.92}$	$\hat{Y}_{0.80}$	$\hat{Y}_{0.76}$	$\hat{Y}_{0.71}$
X1	1	0.95	1	1	1	1	1
X2	1	0.92	0	1	1	1	1
X3	0	0.80	0	0	1	1	1
X4	1	0.76	0	0	0	1	1
X5	1	0.71	0	0	0	0	1

	$\begin{pmatrix} FPR \\ TPR \end{pmatrix}$	$\begin{pmatrix} FPR \\ TPR \end{pmatrix}$	$\begin{pmatrix} FPR \\ TPR \end{pmatrix}$	$\begin{pmatrix} FPR \\ TPR \end{pmatrix}$

Step 2 take threshold

BAD

AUC → 0.50
↓
50%

AUC

↳ For imbalance data → Don't use AUC

↳ $\hat{Y}$-score

↑

ordering

↑

S-Mrat

↑

log-loss

↑

- prob-score

↑

Binary-classification

$$* \log(0.9) * 1 + (0) * \_\_\_$$

$\Downarrow$ $\log(0.9)$

X	Y_i	$\hat{Y} = P$		
$X1$	1	0.9	$\to -\log(0.9) \to 0.0457$	← small
$X2$	1	0.6	$\to -\log(0.6) \to 0.22$	→ Not less
$X3$	0	0.1	$\to$ 0.045 → less	
$X4$	0	0.4		

$$\log(0.1) * 0 + (1-0) * \log(1 - 0.1)$$

$\Downarrow$

$$-\log(0.9)$$

0.22

prob

⇒ probe class (1)

small ↑

good model

$$\text{log-loss} = -\frac{1}{n} \sum_{i=1}^{n} \left\{ \log(p_i) * y_i + (1-y_i) * \log(1-p_i) \right\}$$

Y_i GReal

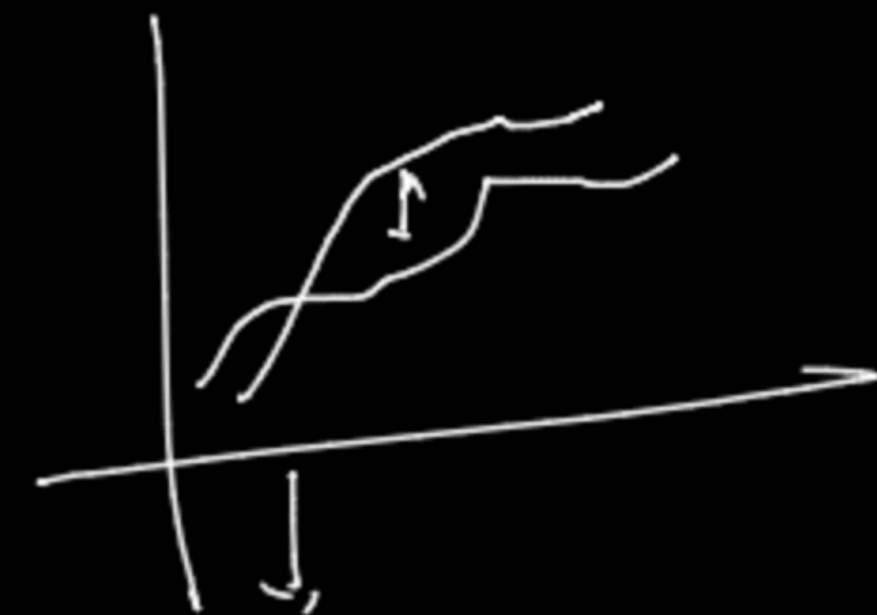
= ↑

X	Y	$\hat{Y}$	e_i
X1	142	165	23
X2	149	150	-
X3	147	201	-
X4	148	199	-

Avg

Regression

R^2 or Coeffⁿ of determination



=> MSE

=> RMSE

=> R^2

=> MAPE

=> Regression

$$\text{error} = (Y_i - \hat{Y}_i)^2$$

$$\text{total error} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

sum of square

$$\sum_{i=1}^{n} (y_i - \bar{y})^2$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$$